FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 16-09-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

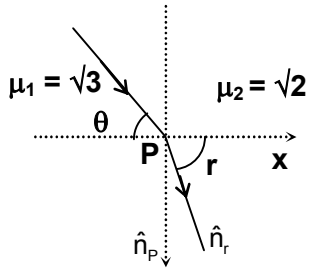
SECTION – A

1. C

Sol. Angle between normal x-axis and incident ray is

$\left(5\sqrt{2}\hat{i}-5\hat{j}-5\hat{k}\right).(\hat{i})=10\cdot1\cdot\cos\theta$

$\cos\theta=\frac{1}{\sqrt{2}}\Rightarrow\theta=45°$



Applying snell's Law at P

$\mu_1\sin\theta=\mu_2\sin r$

$\sqrt{3}\cdot\frac{1}{\sqrt{2}}=\sqrt{2}\sin r$

$r=60°$

The unit vector along the refracted ray

$\hat{n}_r=\cos r\hat{i}+\sin r\hat{n}_P$

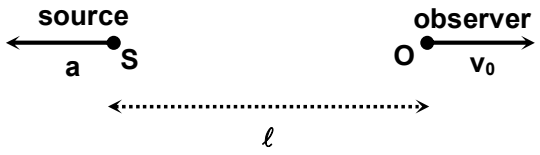
$\hat{n}_r=\cos60°\hat{i}+\sin60°\left(\frac{-\hat{j}-\hat{k}}{\sqrt{2}}\right)$

$\hat{n}_r=\frac{1}{2}\hat{i}+\frac{\sqrt{3}}{2\sqrt{2}}\left(-\hat{j}-\hat{k}\right)$

$\hat{n}_r=\frac{\sqrt{2}\hat{i}-\sqrt{3}\hat{j}-\sqrt{3}\hat{k}}{\sqrt{8}}$

2. B

Sol. Apparent wavelength



$$\lambda' = vT + \frac{1}{2}aT^2$$

Apparent frequency received by the observer

$$f' = \frac{v - v_0}{\lambda'} = \frac{(v - v_0)}{vT + \frac{1}{2}aT^2}$$

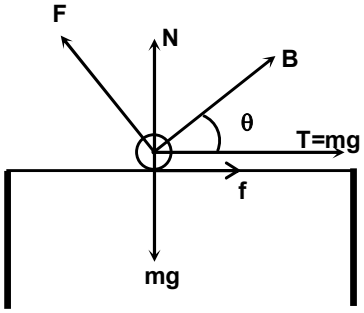
$$= \frac{(v - v_0)}{\frac{v}{f} + \frac{a}{2f^2}} = \frac{2(v - v_0)f^2}{(2vf + a)}$$

3. A

Sol. $$F_{min} = \frac{\mu mg + mg}{\sqrt{1+\mu^2}}$$

$$i\ell B_{min} = \frac{\mu mg + mg}{\sqrt{1+\mu^2}}$$

$$B_{min} = 6 \text{ Tesla}$$



4. C

Sol. $v_0 = A\omega = A\sqrt{\frac{k}{m}} = 5\ell_0\sqrt{\frac{k}{m}}$

$A = 5\ell_0$

$x = A \sin \omega t$

$$3\ell_0 = 5\ell_0 \sin\left(\sqrt{\frac{k}{m}}t_1\right), \quad t_1 = \left(\frac{37^\circ}{180^\circ}\pi\right)\sqrt{\frac{m}{k}}$$

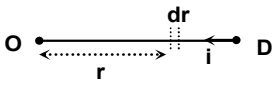
$$4\ell_0 = 5\ell_0 \sin\left(\sqrt{\frac{k}{m}}t_2\right), \quad t_2 = \left(\frac{53^\circ}{180^\circ}\pi\right)\sqrt{\frac{m}{k}}$$

So, time period of oscillation

$$T = 2(t_1 + t_2) = 2\left(\frac{90^\circ}{180^\circ}\pi\sqrt{\frac{m}{k}}\right) = \left(\pi\sqrt{\frac{m}{k}}\right)$$

5. B, C, D

Sol. Taking torque about 'O' at time 't'



$$\tau = \int_0^{\ell}\left[\frac{\frac{1}{2}B\omega\ell^2}{R}\cdot dr \cdot B\right]r$$

$$\tau = \frac{B^2\omega\ell^4}{4R}$$

6. A, B, C, D

Sol. $K.E. = \frac{1}{2}mv^2 \propto \left(\frac{z}{n}\right)^2$

$\Rightarrow v \propto \frac{z}{n}$

$f = \frac{v}{2\pi r} \propto \frac{z^2}{n^3}$

$L = \frac{nh}{2\pi} \propto n$

7. A, B, C, D

Sol. (A) When S_1, S_2 both are open capacity of the system is C_1. Then C_1.

$$C_1 = \frac{\frac{4\pi\varepsilon_0(4R)(2R)}{(4R-2R)}(4\pi\varepsilon_0 6R)}{\frac{4\pi\varepsilon_0(4R)(2R)}{(4R-2R)} + 4\pi\varepsilon_0 6R}$$

$$C_1 = \frac{48\pi\varepsilon_0 R}{5}$$

(B) When S_1 is closed but S_2 is open capacity is C_2. Find C_2.

$$C_2 = \frac{(48\pi\varepsilon_0 R)(24\pi\varepsilon_0 R)}{(48\pi\varepsilon_0 R + 24\pi\varepsilon_0 R)} = 16\pi\varepsilon_0 R$$

(C) When S_2 is closed but S_1 is opened capacity is C_3. Find C_3

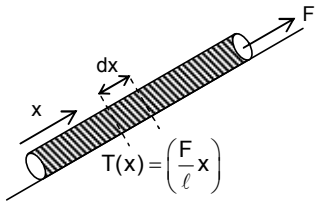
$$C_3 = \frac{(24\pi\varepsilon_0 R)(24\pi\varepsilon_0 R)}{(24\pi\varepsilon_0 R + 24\pi\varepsilon_0 R)} = 12\pi\varepsilon_0 R$$

(D) when both S_1 and S_2 are closed. Capacity is C_4 is. Then C_4 is equal to

$C_4 = 24\ \pi\varepsilon_0 R$

8. A, B, C

Sol. Suppose $d\delta$ is increment in dx.



Then $\int d\delta = \frac{F}{YA\ell}\int_0^{\ell} x\,dx$

$\delta = \frac{F\ell}{2YA}$

$U = \frac{1}{2}kx^2 = \int \frac{1}{2}\left(\frac{YA}{dx}\right)\left(\frac{dx \cdot T}{YA}\right)^2 = \frac{1}{6}\left(\frac{F^2L}{YA}\right)$

9. A, B, C, D

Sol. $\frac{3}{5}T = ma_x$

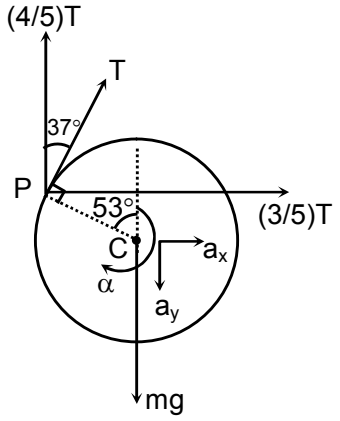
$a_x = \frac{3T}{5m}$ …(i)

$mg - \frac{4}{5}T = ma_y$

$a_y = g - \frac{4T}{5m}$ …(ii)

$\tau_C \Rightarrow TR = \frac{2}{5}mR^2\alpha$

$R\alpha = \frac{5T}{2m}$ …(iii)



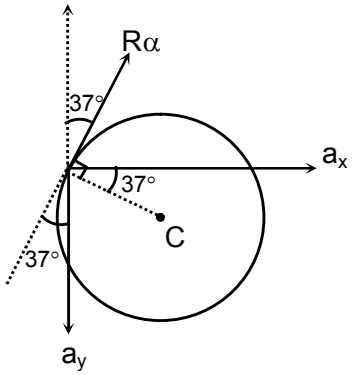
acceleration of point P along the string 'S1' is zero

$R\alpha + a_x \sin 37° = a_y \cos 37°$

$R\alpha + \frac{3}{5}a_x = \frac{4}{5}a_y$ …(iv)

$\frac{5T}{2m} + \frac{3}{5}\left(\frac{3T}{5m}\right) = \frac{4}{5}\left(g - \frac{4T}{5m}\right)$

$T = \frac{8}{35}mg$



10. A, B, C, D

Sol. $C = C_v + \frac{P}{n}\left(\frac{dV}{dT}\right)$

$= \frac{3}{2}R + \left(\frac{nRT}{V \cdot n}\right)\left(\frac{V_0^2}{2VT_0}\right)$

$= \frac{3}{2}R + \frac{RTV_0^2}{2T_0V^2}$

$= \frac{3}{2}R + \frac{R}{2} = \frac{4}{2}R = 2R$

$T = \left(\frac{T_0}{V_0}\right)V^2$

$\frac{dT}{dV} = \frac{T_0}{V_0^2}2V$

$T_f = \left(\frac{T_0}{V_0^2}\right)(2V_0)^2 = 4T_0$

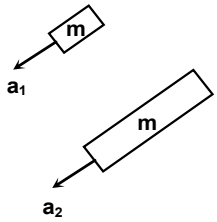
$\Delta Q = nC\Delta T = 1\ (2R)\ (4T_0 - T_0) = 6RT_0$

11. A, B, D

Sol. $\int_{P_1}^{P_5} \vec{B} \cdot \overrightarrow{d\ell} = \left(\frac{45°}{360°}\mu_0 I\right) + \left(\frac{45°}{360°}\mu_0 I\right) + \left(\frac{30°}{360°}\mu_0 I\right) - \left(\frac{15°}{360°}\mu_0 I\right) = \frac{7}{24}\mu_0 I$

12. A, B, C, D

Sol. a_1 and a_2 are acceleration of the block and the plank with respect to ground



$$a_1 = \frac{4}{5}g - \frac{1}{4}\left(\frac{3}{5}g\right) = \frac{13}{2}\ m/s^2$$

$$a_2 = \frac{4}{5}g + \frac{1}{4}\left(\frac{3}{5}g\right) - 2\left(\frac{1}{2}\right)\left(\frac{3}{5}g\right) = \frac{7}{2}\ m/s^2$$

$a_{12} = a_1 - a_2 = \frac{13}{2} - \frac{7}{2} = 3\ m/s^2$

$$\ell = ut + \frac{1}{2}at^2$$

$$1.8 = \frac{1}{2}(3)t^2$$

$t = \sqrt{1.2}$ sec

$$d_1 = \frac{1}{2}a_1t^2 = \frac{1}{2}\times\frac{13}{2}\times 1.2 = 3.9\ m$$

SECTION – C

13. 00024.89

Sol. Error $= \left(\frac{270^\circ}{360^\circ}\times 100\right)\times\frac{1\,mm}{100} = 0.75\,mm$

Measured value = 24 + 0.26 –0.12 + 0.75 = 24.89 mm

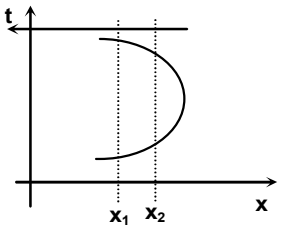
14. 00001.20

Sol. Charge particle lies at centroid of equilateral pyramid.

$$\phi = \frac{\left(\frac{q}{\varepsilon_0}\right)}{12} = \frac{q}{12\varepsilon_0}$$

15. 00005.10

Sol. $$a_0 = \frac{8(x_2 - x_1)}{(t_1^2 - t_2^2)} = \frac{8\times 7.65}{(16-4)} = 5.10$$



16. 00012.50

Sol. Impulse, $\int Fdt = m\sqrt{2gh}$

$\int(i\ell B)dt = m\sqrt{2gh}$ …(i)

$\int idt = \Delta q$ …(ii)

$$\Delta q = \frac{m\sqrt{2gh}}{B\ell} = \frac{20\sqrt{2\times 10\times 5}}{2\times 8} = 12.50\ \text{Coulomb}$$

17. 00000.80

Sol. $I_{C.B.F.} = K\left[A_1^2 + A_2^2 + 2A_1A_2\cos\phi\right] = I_0$

$KA^2 = \frac{I_0}{16}$, ($A_1 = A$, $A_2 = 3A$ and $\phi = 0°$)

$I_\theta = \frac{16}{25}K\left[A_1^2 + A_2^2 + 2A_1A_2\cos\phi\right] = I$

$\phi = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} \times d\sin\theta = \frac{2\pi}{\lambda} \times d\frac{3}{5}$

So, $\phi = \pi$

Hence, $I = \frac{I_0}{4} \times \frac{16}{25} = \frac{5 \times 16}{100} = 0.80$ W/m^2

18. 00014.80

Sol. $\frac{dN}{dt} = r - \lambda N$

$\int_{N_0}^{N} \frac{dN}{r - \lambda N} = \int_0^t dt$

$\frac{dN}{dt} = r - \lambda N = (r - \lambda N_0)e^{-\lambda t}$

$= \left(10^4 - \frac{10^5}{20}\right)e^{-t/20} = \frac{5000}{7.4} = \left(\frac{10^4}{14.8}\right)$

Chemistry

PART – II

SECTION – A

19. C

Sol. Thermal stability decreases with increase in bond length.

20. C

Sol. $\Delta T_b = \text{molality}.\ K_b$

$\Delta T_f = \text{molality}.\ K_f$

$\therefore \frac{\Delta T_b}{\Delta T_f} = \frac{K_b}{K_f}, \qquad \Delta T_f = \Delta T_b . \frac{K_f}{K_b}$

$\therefore \Delta T_f = 1.56 \times \frac{1.86}{0.52} = 5.58$

21. A

Sol. $\sqrt{\frac{8RT_1}{\pi M}} = \sqrt{\frac{2RT_2}{M}}$

$\therefore \frac{T_1}{T_2} = \frac{2\pi}{8} = \frac{3.14}{4} = 0.785$

22. D

Sol. $r = \frac{PA}{\sqrt{2\pi MRT}}$

$r_{He} : r_{N_2} : r_{O_2} = \frac{1}{\sqrt{4}} : \frac{1}{\sqrt{28}} : \frac{1}{\sqrt{32}}$

$\therefore P_{He} < P_{N_2} < P_{O_2}$

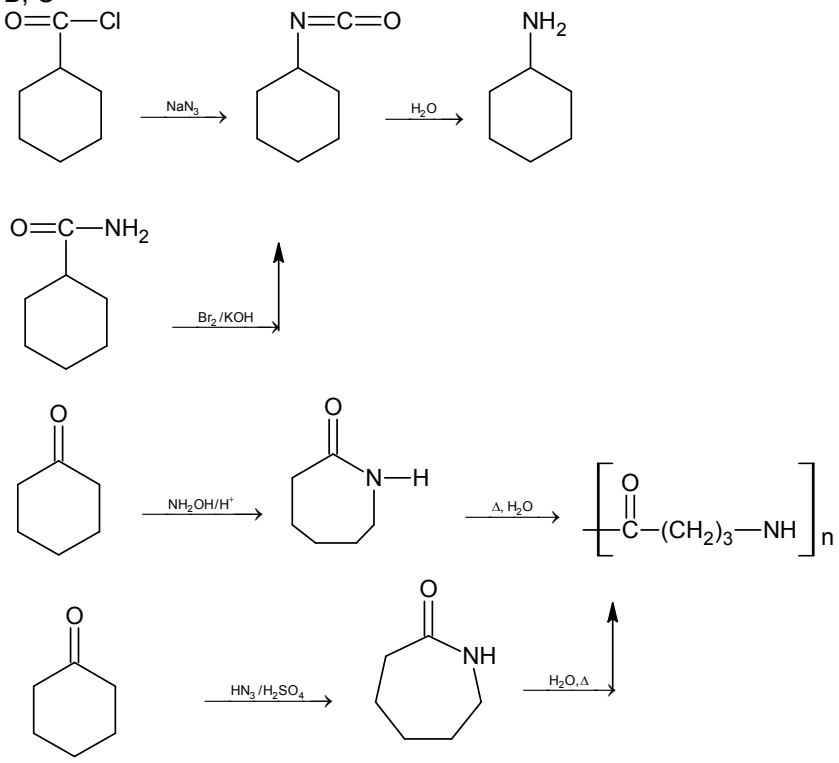
23. A, B, C

Sol. Facts.

24. C

25. B, C

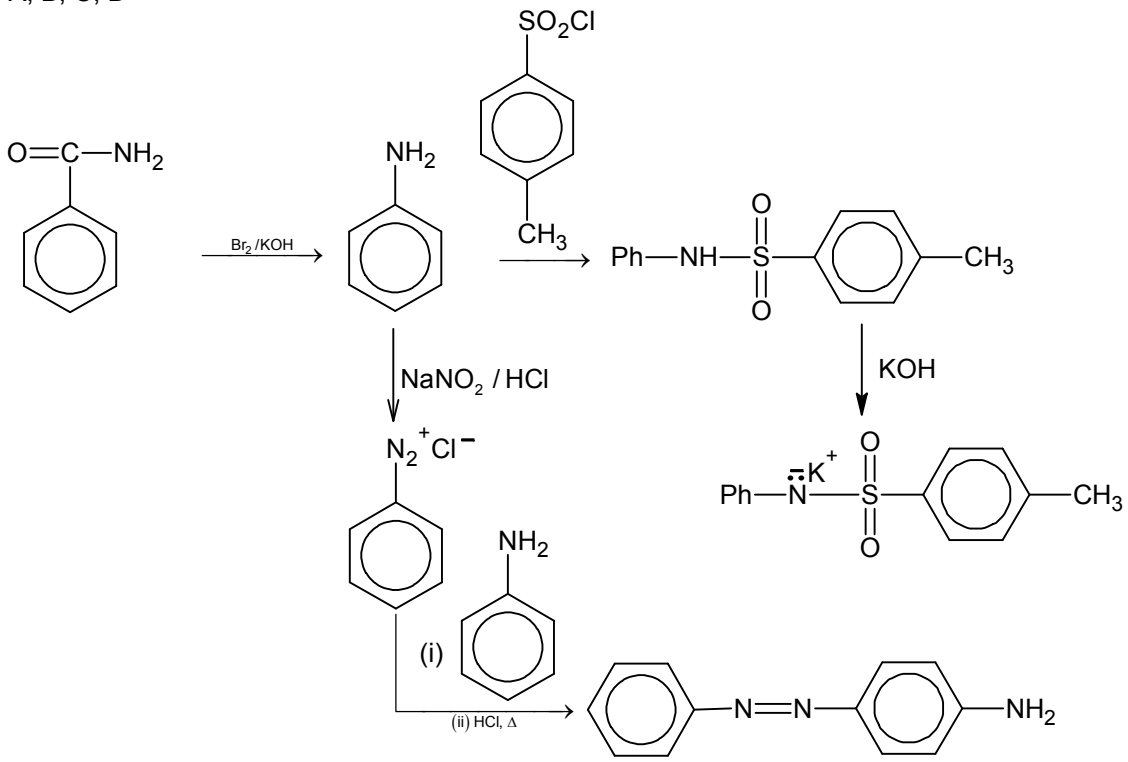
Sol.



26. A, D

27. A, B, C, D

Sol.



28. A, B, C

Sol. $2NH_3 + 2H_2O \rightleftharpoons 2NH_4^+ + 2\bar{O}H$

$H_2S \rightleftharpoons 2H^+ + S^{2-}$

$2H^+ + 2\bar{O}H \longrightarrow 2H_2O$

Hence, conc. Of S^{2-} is large.

$\therefore$ Both IInd and IV group cation will precipitate as S^{2-} salt.

29. A, B, D

Sol. Cinnabar $\rightarrow$ HgS
Galena $\rightarrow$ PbS
Chalcocite $\rightarrow$ Cu_2S
Bauxite $\rightarrow$ $Al_2O_3.2H_2O$

Sulphide ores can be concentrated by this method.

30. A, B, C, D

Sol. $BiCl_3 \xrightarrow{NaOH} Bi(OH)_3 + NaCl$

$\underset{(P)}{Bi(OH)_3} \xrightarrow[ii)\ KI]{i)\ conc.\ HCl} \underset{\text{Black (U)}}{BiI_3 \downarrow} \xrightarrow{KI} \underset{\text{Orange (V)}}{K[BiI_4]}$

$\underset{(Q)}{NaCl} \xrightarrow[H_2SO_4]{K_2Cr_2O_7} \underset{\text{Red gas (R)}}{CrO_2Cl_2 \uparrow} \xrightarrow{NaOH} \underset{\text{Yellow solution (S)}}{Na_2CrO_4} \xrightarrow{AgNO_3} \underset{\text{Brick red ppt. (T)}}{Ag_2CrO_4 \downarrow}$

SECTION – C

31. 00060.25

Sol. $K_H = P_A/Y_A$

Partial pressure of O_2 = 0.21 atm = 21278.2 Pa.

$P_{N_2} = 0.79$ atm $= 80046.8$ Pa

X_{O_2} in dissolved water $= \dfrac{21278.2}{2.03 \times 10^8} = 10.4819 \times 10^{-5}$

X_{N_2} dissolved in water $= \dfrac{80046.8}{5.07 \times 10^8} = 15.7883 \times 10^{-5}$

$\dfrac{X_{O_2}}{X_{N_2}} = 0.6639 = 0.66$

% of $O_2 = \dfrac{0.66}{1.66} \times 100 = 39.75$

% of N_2 = 60.25

32. 00061.83

Sol. Mass of dissolved naphthalene = 0.624 gm

Mass of solvent (CCl_4) = 0.060 kg

Molality = $= \dfrac{0.624 / 128.2}{0.060} = 0.08$

$\Delta T_b = K_b m$

$\therefore K_b = \dfrac{0.42}{0.08} = 5.25$

For unknown solution

$0.75 = 5.25 \times \dfrac{0.53}{M \times 0.06}$

$M = 61.8333 \approx 61.83$

33. 00001.53

Sol. $M_{mix} = \frac{(2\times 28)+(3\times 4)}{5} = 13.6$

$\frac{r_{mix}}{r_{O_2}} = \frac{5}{5}\sqrt{\frac{32}{13.6}} = 1.5339 \approx 1.53$

34. 00023.00

Sol. $V_2 = \frac{P_1V_1}{P_2} = \frac{1\times 10}{1} = 10\ m^3$

$n = \frac{P_1V_1}{RT} = \frac{10\times 1\times 10^3}{(0.0821)(273.15)} = 445.9186$

≈ 445.92

$W = -2.303\ nRT\log\ \frac{P_1}{P_2}$

$= -2.303 \times (445.82)\ (0.0821)\ (273.15)\log 10$

$= -23002.018\ J = -23.00\ kJ$

35. 00001.15

Sol. $LiCl.3NH_3(s) \rightleftharpoons LiCl.NH_3(s) + 2NH_3(g)$

$LiCl.NH_3(s) + 2NH_3(g) \rightleftharpoons LiCl.3NH_3(s),\ K'_P = \frac{1}{16}\ atm^{-2}$

Initially	0.2	a	–
Finally	0	$a - 0.4$	0.2

$K'_P == \frac{1}{P^2_{NH_3}} = \frac{1}{16}$

$\therefore P_{NH_3} = 4$ atm at equilibrium

PV = nRT

$4 \times 5 = n \times 0.0821 \times 323$

$n = 0.75419 \approx 0.75$ moles at equilibrium.

$\therefore$ Initial moles of $NH_3 = 0.75 + 0.4 = 1.15$.

36. 00209.18

Sol. $n = PV/RT = \frac{(1)(200\times 10^{-3})}{(0.0821)(273)} = 8.92\times 10^{-3}$

Number of Xe atom = $8.92 \times 10^{-3} \times 6.02 \times 10^{23}$

$= 53.70 \times 10^{20}$

Volume occupied by atoms $= \left(\frac{4}{3}\pi r^3\right) \times 53.70 \times 10^{20}$

$= 418.36 \times 10^{-7}$

% of Volume occupied by He atoms = $= \frac{418.36\times 10^{-7}}{200} \times 100$

$= 209.18$

Mathematics

PART – III

SECTION – A

37. C

Sol. Let A(0, a) and B(b, 2b)

Coordinate of P is $\left(\frac{2a}{5}, \frac{4a}{5}\right)$

Coordinate of Q is (0, 2b)

Equation of AB is $y - a - \left(2 - \frac{a}{b}\right)x = 0$

$\because$ It passes through (1, 3) hence $a = \frac{b}{b-1}$

Also equation of PQ is $y - 2b = \left(2 - \frac{5b}{a}\right)x$

i.e., $y - 7x + b(x - 2) = 0$

$\therefore$ Fixed point is $\left(\frac{2}{5}, \frac{14}{5}\right)$

38. B

Sol. $\int_0^{\pi/2} \frac{\sin 40x}{\sin 5x} = 2\int_0^{\pi/2} [\cos 5x + \cos 15x + \cos 25x + \cos 35x]dx = \frac{2}{5}\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}\right)$

39. C

Sol. $e^{2x} f(x) = -\frac{1}{2}\tan x + e^{2x}\int_0^x e^{-2t}\tan t dt$

$e^{2x}\left(f'(x) + 2f(x)\right) = -\frac{1}{2}\sec^2 x + e^{2x}e^{-2x}\tan x + 2\int_0^x e^{-2t}\tan t\,dt \cdot e^{2x}$

$\Rightarrow e^{2x}\left(f'(x) + 2f(x)\right) = -\frac{1}{2}\sec^2 x + \tan x + 2\left(e^{2x}f(x) + \frac{1}{2}\tan x\right)$

$\Rightarrow e^{2x}f'(x) = 2\tan x - \frac{1}{2}\sec^2 x \Rightarrow f'(x) = e^{-2x}\left(2\tan x - \frac{1}{2}\sec^2 x\right)$

40. B

Sol. Let $\angle ABD = \theta$

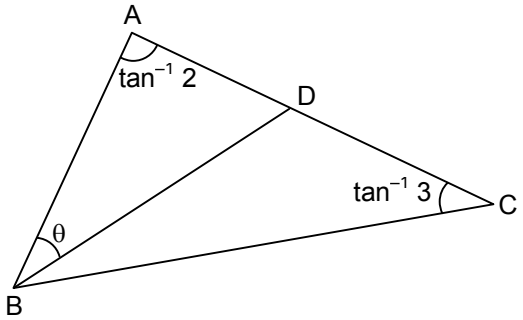
$\because \Delta ABD = \frac{9}{2}\Delta BCD$

$\frac{1}{2}AB\,BD\sin\theta = \frac{9}{2}\cdot\frac{1}{2}BC\,BD\sin\left(\frac{\pi}{4} - \theta\right)$

$\Rightarrow \frac{\sin\left(\tan^{-1}3\right)}{\sin\left(\tan^{-1}2\right)} = \frac{9}{2\sqrt{2}}\left(\frac{\cos\theta - \sin\theta}{\sin\theta}\right)$

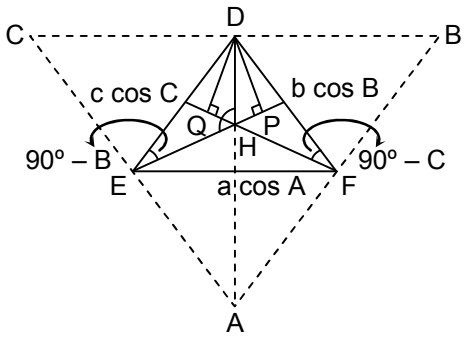
$(\cot\theta - 1)\frac{9}{2\sqrt{2}} = \frac{3}{2\sqrt{2}}$

$\Rightarrow \cot\theta - 1 = \frac{1}{3} \Rightarrow \cot\theta = \frac{4}{3}$



41. A, C

Sol. $\sin(90^\circ - B) = \dfrac{DP}{DE}$



$\Rightarrow$ DP = c cos C·cos B

Also applying sine rule in ΔDEH

$$\frac{c\cos C}{\sin(180^\circ - C)} = \frac{EH}{\sin(90^\circ - A)}$$

$$\Rightarrow EH = \frac{c\cos C\cos A}{\sin C}$$

$$\therefore \frac{DP}{EH} = \frac{\sin C\cos B}{\cos A}$$

Similarly $\dfrac{DQ}{FH} = \dfrac{\sin B\cos C}{\cos A}$ $\therefore$ $\dfrac{DP}{EH} + \dfrac{DQ}{FH} = \tan A$

42. A, D

Sol. $I_1 = \int_0^{\frac{\pi}{200}} \sin\left(2\pi\cos^2 100x\right)dx$

$$I_1 = \int_0^{\frac{\pi}{200}} \sin\left(2\pi\sin^2 100x\right)dx$$

$$2I_1 = \int_0^{\frac{\pi}{200}} 2\sin\pi\cdot\cos\left(\pi\cos 200x\right)dx = 0$$

$$I_2 = \int_0^{\frac{\pi}{200}} \sin\left(\pi\sin^2 100x\right)dx = \int_0^{\frac{\pi}{200}} \sin\left(\pi\cos^2 100x\right)dx$$

$$2I_2 = \int_0^{\frac{\pi}{200}} 2\cos\left(\frac{\pi}{2}\cos 200x\right)dx$$

$$I_2 = 2\int_0^{\frac{\pi}{400}} \cos\left(\frac{\pi}{2}\cos 200x\right)dx$$

Let 2x = t, $\displaystyle\int_0^{\frac{\pi}{200}} \cos\left(\frac{\pi}{2}\cos 100t\right)dt = \int_0^{\frac{\pi}{200}} \sin\left(\pi\sin^2 50t\right)dt$

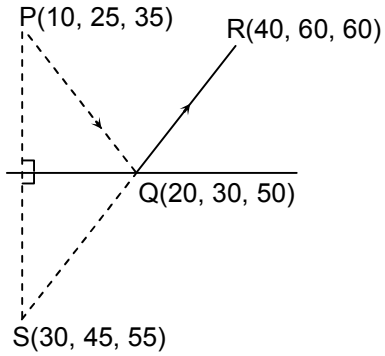
43. B, C

Sol.

$$f(x) = \begin{cases} 2\pi + 4\tan^{-1}\dfrac{x}{2} & ; \quad x < -2 \\ \pi & ; \quad -2 \le x < 0 \\ \pi - 4\tan^{-1}\dfrac{x}{2} & ; \quad 0 \le x \le 2 \\ 0 & ; \quad x > 2 \end{cases}$$

44. B, C, D

Sol. Image of P(10, 25, 35) about the plane x + y + z = 100 is S(30, 45, 55)



Here, equation of reflected ray is

$$\frac{x-40}{2}=\frac{y-60}{3}=\frac{z-60}{1}=r$$

Also, point Q is (20, 30, 50)

$\therefore$ Equation of incident ray is $\frac{x-10}{2}=\frac{y-25}{1}=\frac{z-35}{3}$

Equation of plane containing the line PQ and RQ is

$$\begin{vmatrix} x-10 & y-25 & z-35 \\ 2 & 1 & 3 \\ 2 & 3 & 1 \end{vmatrix}=0 \text{ i.e., } 2x-y-z+40=0$$

45. A, C

Sol. $f'(C_1)=\frac{f(8)-f(0)}{8}=\frac{1}{2}$

$f(C_2)=\frac{f(4)+f(8)}{2}=3$

Let $g(x)=\int_0^{x^{3/2}} f(t)dt$

$\Rightarrow g'(x)=\frac{3}{2}\sqrt{x}\, f\left(x^{3/2}\right)$

$g'(\alpha)+g'(\beta)=\frac{g(4)-g(0)}{2}$

$\Rightarrow 3\left[\alpha f\left(\alpha\sqrt{\alpha}\right)+\beta f\left(\beta\sqrt{\beta}\right)\right]=\int_0^8 f(t)dt$

Let $g(x)=\int_0^{x^2} f(t)dt$

$g'(x) = 2xf(x^2)$

$g'(\alpha)=\frac{g(2)-g(0)}{2}$

$g'(\beta)=\frac{g(4)-g(2)}{2}$

$\Rightarrow g'(\alpha)+g'(\beta)=\frac{g(4)-g(0)}{2}$

46. A, B, C, D

Sol. $f(\alpha)=\frac{\alpha}{\sqrt{1-\alpha^2}}$

47. A, C, D

Sol. Let removed numbers are 2r – 1, 2r + 3, 2r + 7

$$\frac{n^2-3(2r+3)}{n-3}=\frac{105}{2}$$

$\Rightarrow r = \dfrac{2n^2 - 105n + 297}{12}$

$\because\ 1 \le \dfrac{2n^2 - 105n + 297}{12} \le n - 4$

$\Rightarrow 50 \le n \le 55$

Possible values of n are 51, 53, 55

For n = 51, r = 12 $\therefore$ Removed numbers are 23, 27, 31

48. A, B, C

Sol. $(A - 2I)^4 = 5A \Rightarrow \text{Det}(A - 2I)^4 = \text{Det}(5A) \Rightarrow \text{Det}(A - 2I) = \pm 5$

$\therefore$ Det B = 5. Also, $I - 2A^{-1} = A^{-1}B$, $\text{Det}(I - 2A^{-1}) = \text{Det}(A^{-1}B) = 1$

$\Rightarrow \text{Det}(\text{Adj}(I - 2A^{-1})) = 1,\ \text{Det}\left(\dfrac{B}{5}\right)^{-1} = \dfrac{1}{\text{Det}\left(\dfrac{B}{5}\right)} = 25$

$\Rightarrow \text{Det}\left(\text{Adj}\left(\dfrac{B}{5}\right)\right)^{-1} = 25^2 = 625$

Also, $I + 2B^{-1} = AB^{-1} \Rightarrow \text{Det}(I + 2B^{-1}) = 1$

SECTION – C

49. 00054.00

Sol. $\hat{p} = \dfrac{\hat{a} + \hat{b}}{\left|\hat{a} + \hat{b}\right|} = \dfrac{\hat{a} + \hat{b}}{\sqrt{3}}$

Similarly $\hat{q} = \dfrac{\hat{b} + \hat{c}}{\sqrt{3}}$ and $\hat{r} = \dfrac{\hat{c} + \hat{a}}{\sqrt{3}}$

Volume of regular tetrahedron = $\dfrac{\sqrt{3}}{2\sqrt{2}}$

$\left[\hat{p} \times \hat{q}\quad \hat{q} \times \hat{r}\quad \hat{r} \times \hat{p}\right] = \left[\hat{p}\ \hat{q}\ \hat{r}\right] = \left[\dfrac{\hat{a} + \hat{b}}{\sqrt{3}}\ \dfrac{\hat{b} + \hat{c}}{\sqrt{3}}\ \dfrac{\hat{c} + \hat{a}}{\sqrt{3}}\right]^2 = \dfrac{4}{27} \times \left[\hat{a}\ \hat{b}\ \hat{c}\right]^2 = 2$

50. 01854.00

Sol. $a - 1 + ar^2 = 2ar$

$\Rightarrow (r - 1)^2 = \dfrac{1}{a} \Rightarrow r = 1 \pm \dfrac{1}{\sqrt{a}}$; $a = k^2,\ k \in I$

If a = 4, then $r = \dfrac{3}{2}, \dfrac{1}{2}$

$\therefore$ abc = 469 or 421

If a = 9, then $r = \dfrac{2}{3}, \dfrac{4}{3}$

$\therefore$ abc = 964; $r \ne \dfrac{4}{3}$

51. 00008.00

Sol. $\dfrac{a^{20}b + b^{20}a - (a + b)\left(a^{19}b + b^{19}a\right)}{a^{18}b + b^{18}a} = -ab$

52. 00005.00

Sol. $f(x)=\frac{10}{9}\frac{x^5}{1+x^4}$

$f(x)=\frac{5}{9}$ at x = 1

Required area $A=\frac{5}{9}-\int_0^1\frac{10}{9}\frac{x^5}{1+x^4}dx = \frac{5\pi}{36}$

53. 00006.00

Sol. For f(x) to be continuous at x = 2 – k or 2 + k

$\frac{\sin\alpha}{4}+k^2=\frac{\cos\alpha}{4}+k$ ..... (1)

$f'(2-k)^- = -1$, $f'(2-k)^+ = -2k$

$\because$ f(x) is differentiable

$\therefore f'(2+k)^+ = f'(2+k)^- \Rightarrow k = \frac{1}{2}$

54. 00002.00

Sol. $20\left(1+x+x^2\right)^{20}(1+2x)=\sum_{k=1}^{40}\left(ka_kx^{k-1}\right)\left(1+x+x^2\right)$

Equating coefficient of x^{10} on both sides

$20\left[a_{10}+2a_9\right]=11a_{11}+10a_{10}+9a_9$

$\Rightarrow 11a_{11} = 10a_{10} + 31a_9$

$\Rightarrow \frac{\gamma-\alpha}{\beta}=2$